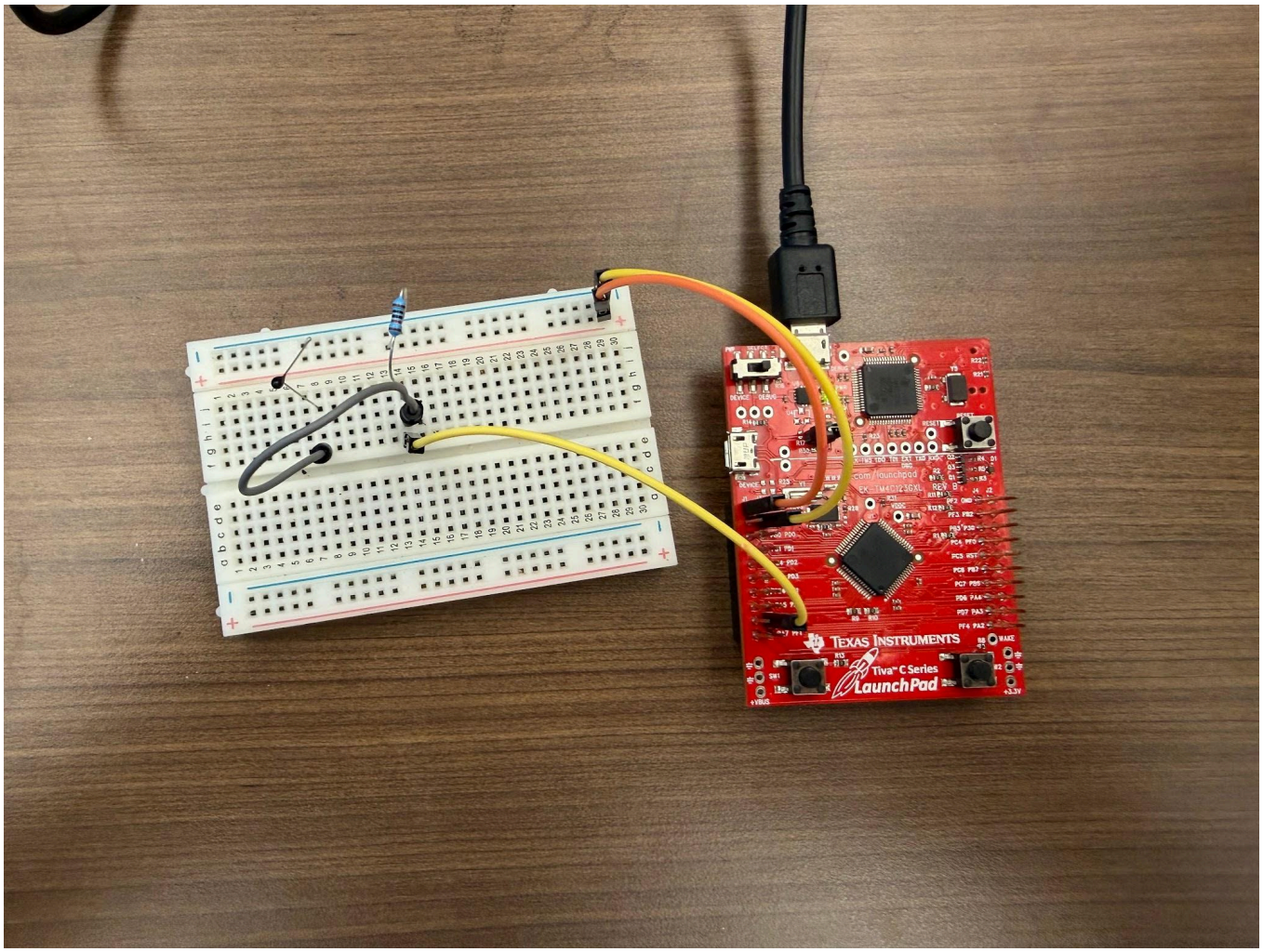
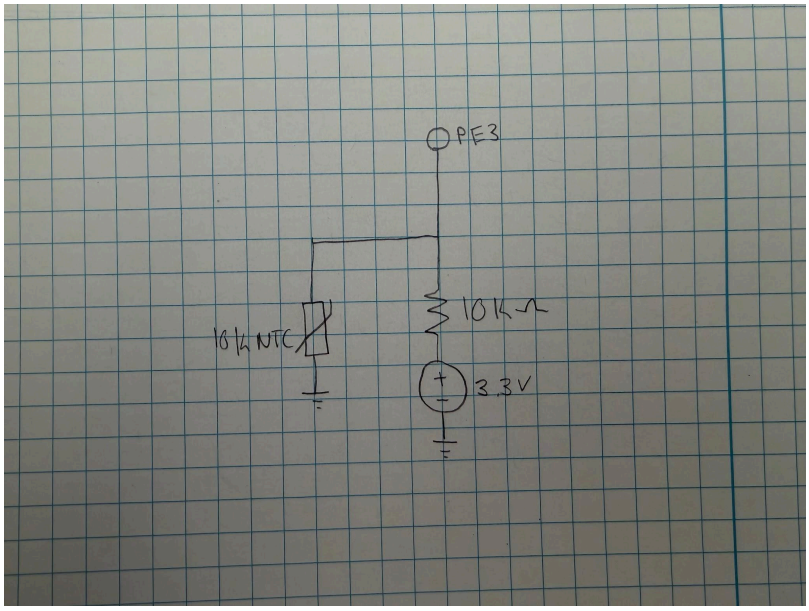
Lab 4: Report

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R1: Prototype and Circuit Diagram





Fixed Resistor Value: **10 kOhm**

We chose a 10 kΩ fixed resistor because it matches the thermistor's resistance at room temperature 10 kΩ at 25 °C. When the two resistors are equal, the voltage at PE3 is exactly half of 3.3 V, about 1.65 V, which is right in the middle of what the ADC can read. The thermistor's resistance was measured with a multimeter.

R2: Difference Between Temperature Readings

The two sensors did not read the same value. The internal sensor read about 15 °C, while the thermistor read about 20-21 °C, a difference of roughly 5 °C. The thermistor's reading was more accurate, since its reading comes from a simple voltage divider and the manufacturer's published constants. The internal sensor is a coarse sensor built into the chip and isn't meant to be precise.

The most likely source of the difference is thermal contact with the board. The internal sensor measures the temperature of the chip itself, which is attached to the board and the USB-powered debug circuitry. The thermistor sits on the breadboard on its own wires and mostly measures the air around it. Because they are in different spots and make contact with different surfaces, they can read different temperatures even in the same room.

Contributions: Both students worked across each part, but this is a list of the majority work done.

Paul- Part 1 code and circuit, report

Benjamin- Part 2 code and circuit, circuit diagrams